

NIKAN KHADKA

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Full-stack JavaScript/TypeScript developer with 3years of production software development and AI/ML project experience, currently completing a Master of Information Technology in Artificial Intelligence at Charles Darwin University, expected Oct 2026. Experienced across Next.js, Node.js, Express.js, NestJS, PostgreSQL, MongoDB, REST APIs, Docker, CI/CD, Playwright and Jest. Built SaaS automation, LMS modules, AI-assisted workflows, RAG/LLM systems with Azure AI Foundry, customer-feedback analytics/NLP chatbot pipelines and a multi-vendor rental platform. Seeking junior, graduate, full-stack, QA automation or GenAI Engineer roles.

TECHNICAL SKILLS

Programming: JavaScript, TypeScript, Python, Java

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, Zustand, React query,Figma

Backend: Node.js, Express.js, Fastify, NestJS, Python FastAPI REST APIs, JWT, OAuth, WebSockets, Socket.IO, Swagger/OpenAPI

Databases: PostgreSQL, MongoDB, MySQL, Supabase

Testing & QA: Playwright, Jest, Postman, unit testing, E2E testing, regression testing, manual QA, test fixtures

DevOps & Tools: Git, GitHub, GitLab, Docker, GitHub Actions, GitLab CI/CD, Vercel, Render, AWS/EC2 basics, Linux, Azure.

AI/ML & Productivity: NLP, Azure AI Foundry, RAG workflows, LLM systems, multi-agent workflows, AI prompt engineering, Cursor, Claude, Codex

EXPERIENCE

AI Software Engineer - Paypipe, Sydney, Australia

Feb 2026 - Present

Tech: Next.js, TypeScript, Node.js/Express, PostgreSQL, Azure AI Foundry, AI prompting, SaaS, RAG, LLM and multi-agent workflows

- Built **AI-assisted workflows** for a trade-services SaaS platform, integrating LLM-powered assistants into client-facing chat for job scoping, quotation drafting and milestone-based offer generation.
- Engineered **RAG-style conversation grounding** so AI outputs could use active customer-provider chat history as context, improving relevance for job scopes, pricing breakdowns and service-provider responses.
- Designed a practical **agentic offer workflow** where structured inputs, PDF summaries, voice/audio translation and prompt-controlled AI responses work together to reduce manual quotation effort.
- Improved frontend business workflow reliability by adding human edit/confirmation steps around AI-generated outputs, keeping providers in control before quotes or milestone offers are sent to clients.
- Supported a staging product used by about **20 traders/service providers and 50 clients**, helping reduce manual messaging and quotation preparation effort by an estimated **70%+** through AI-assisted concierge workflows.

Software Developer- Eightbit / Dzangolab,Nepal

Sep 2023 - Mar 2025

Tech: TypeScript, React, Fastify, PostgreSQL, Socket.IO, Docker, GitLab, Playwright, Jest, Make.com

- Developed full-stack SaaS features across **Dzango-HQ Legal, Dzango LMS and 12deg** using React, Fastify/Node.js, PostgreSQL, REST APIs, Docker and GitLab workflows.
- Dzango-HQ Legal: built UI/API/automation for contracts/documents, leave requests, payslips, calendar, performance reviews, bank/SSF details and approvals; used **Make.com** notifications for document status changes, reducing email/paper workflow steps by **50%+ for 20+ staff**.
- Dzango LMS: modelled and built major modules for courses, skills, students, companies, invitations, teacher assignments, grading, certifications and markdown resources for onboarding/client learning workflows.
- **Improved LMS onboarding visibility** by replacing untracked resource sharing with structured progress, submissions and grading data; personal onboarding testing showed **completion in less than half the usual time**.
- 12deg: built a collaborative markdown editor from scratch with Milkdown/ProseMirror and Socket.IO; contributed release-dashboard graphing with Chart.js, service-dependency visualisation, test fixtures and UI fixes in an 8-developer full-stack team.
- Implemented **GitLab/Docker event-driven release workflows** including branch-to-release mapping, service deploy/stop controls, workspace service status updates, webhook status sync and release graph updates; **tested 5+ branches with 30+ service instances**.
- Wrote around **20 Playwright E2E tests for authentication and release workflows**; added test database fixtures and supported CI/test environment setup, production YAML/env fixes and SMTP configuration tasks.

Full-Stack Developer- Softechtonic, Naxal, Kathmandu

Feb 2023 - Jul 2023

Tech: Node.js, Express-TS, MongoDB, JWT/OAuth, Swagger/OpenAPI, Jest, Postman

- Built backend/admin web features for Doctor/Medicine Management and Codyn timer course-management systems using Node.js, Express-TS, MongoDB, JWT/OAuth and REST APIs.
- Designed schemas and implemented **40+ REST APIs** covering CRUD, validation, authentication and course/healthcare workflows; documented APIs with Swagger/OpenAPI for frontend/mobile integration.
- Contributed to Codyn timer features used in Herald College/testing contexts and worked with another developer on early migration toward a more type-safe NestJS-based architecture.

PROJECTS

MeroGhar: Multi-vendor Rental Management Platform

Feb 2023- 2026

Rebuilt and optimised in 2025-2026 | Live: <https://fyp-pi-sand.vercel.app/Home> | GitHub: github.com/Nikankhadka/fyp

- Built an Airbnb-style rental platform for Nepal with tenant/owner/admin roles, property listings, dashboards, booking, check-in/check-out, real KYC integration, phone/email verification, social login, payments, billing, notifications and verified reviews.
- Revamped the 3-year-old project into a maintainable monorepo with Docker, GitHub Actions CI/CD, seed data/fixtures, documentation and around 15 Playwright tests.
- Optimised production performance: **Lighthouse score 57 to 89, bundle/package size down 20%+** and **API response performance up about 20%** after dependency cleanup, query optimisation and Vercel analytics review.

RestaurantXpert: Aspect Sentiment Analysis Chatbot

2026

Built **RestaurantXpert**, a restaurant review intelligence chatbot that turns unstructured guest reviews into actionable insights across food, service, price and ambience, helping operators identify what customers praise, what causes complaints and where to prioritise improvements.

- Trained an **aspect-based sentiment analysis** pipeline using TF-IDF, SMOTE and Logistic Regression on SemEval-2014 data; achieved **70.99% accuracy** and **0.715 weighted F1** for aspect-level trend detection without GPU or paid LLM dependency.
- Designed a pre-aggregated **knowledge base from 3,693 annotations** to answer business questions instantly, such as which categories attract negative feedback, whether price perception is positive or poor, and where service issues appear most often.
- Added an optional **OpenRouter/GLM-4.5-Air LLM wrapper** with deterministic fallback and guardrails, keeping business-facing answers natural while still grounded in pre-computed review statistics.

EDUCATION

Charles Darwin University, Sydney	M.I.T. - Artificial Intelligence	Mar 2025 - Oct 2026
University of Wolverhampton / Herald College Kathmandu	BSc (Hons) Computer Science	Graduated Nov 2023

AWARDS AND LEADERSHIP

- 2nd Place, Herald Expo/Hackathon (Jul 2022): built a disease-prediction prototype using MERN and a Python dataset model to predict likely conditions from symptom inputs.
- Mentor, Herald Developer Corps: mentored 40+ junior students in web development and supported an LMS project **Codyn timer** with real users under senior academic guidance.
- Coding Workshop Facilitator, Reliance Public School (May 2024): delivered 8 beginner sessions for grades 6-10 using CSS games and practical web examples.
- Highest achiever in Java coursework: scored 96% while building a Java Swing/MySQL course-management system.